

COUNTEREXAMPLES TO CONJECTURE 5.4 OF PARISI–SPAHIU–SKANDERA–WANG

ABSTRACT. Parisi, Spahiu, Skandera, and Wang conjectured that if a permutation w contains the pattern 45312, then the modified signless Kazhdan–Lusztig basis element $\tilde{C}_w$ has no reversal factorization. We give exact identities in $H_6(q)$ showing that seven such permutations admit reversal factorizations, and we prove that these are exactly the factorable 45312-containing elements of $\mathfrak{S}_6$. The other nineteen cases are excluded by the obstruction theorems of Parisi–Spahiu–Skandera–Wang. Larson’s earlier small-resolution construction for 463152, together with the framework of Agrawal and Sotirov, gives a prior geometric route to an associated Kazhdan–Lusztig factorization. The present note gives direct reversal-factorization certificates in the normalization of the conjecture and the complete $\mathfrak{S}_6$ classification.

1. THE COUNTEREXAMPLES

Let $H_n(q)$ be the type- A Hecke algebra over $\mathbb{Z}[q]$ with basis $\{T_w : w \in \mathfrak{S}_n\}$ and multiplication

$$T_x T_{s_i} = \begin{cases} T_{xs_i}, & \ell(xs_i) > \ell(x), \\ qT_{xs_i} + (q-1)T_x, & \ell(xs_i) < \ell(x). \end{cases}$$

We use the modified signless Kazhdan–Lusztig basis [3]

$$\tilde{C}_w = \sum_{v \leq w} P_{v,w}(q) T_v.$$

For $1 \leq a < b \leq n$, let $s[a, b]$ be the longest element of the standard parabolic subgroup generated by $s_a, \dots, s_{b-1}$, and put

$$C_{ab} := \tilde{C}_{s[a,b]}.$$

For a sequence of intervals $\mathcal{A} = ([a_1, b_1], \dots, [a_m, b_m])$, define

$$I_{i,j} = [a_i, b_i] \cap [a_j, b_j] \setminus \bigcup_{i < k < j} [a_k, b_k], \quad f_{\mathcal{A}}(q) = \prod_{i < j} [|I_{i,j}|]_q!,$$

where $[r]_q! = \prod_{k=1}^r (1 + q + \dots + q^{k-1})$. Following [5, Sec. 3], a *reversal factorization* of $\tilde{C}_w$ is an identity

$$f_{\mathcal{A}}(q) \tilde{C}_w = C_{a_1 b_1} \cdots C_{a_m b_m}.$$

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TABLE 1. The seven counterexamples. In every row the displayed interval sequence satisfies $f_{\mathcal{A}}(q) = (1 + q)^2$.

w	positions of 45312	factorization in $H_6(q)$
463152	1, 2, 3, 4, 6	$(1 + q)^2 \tilde{C}_{463152} = C_{24}C_{13}C_{46}C_{24}$
526413	1, 3, 4, 5, 6	$(1 + q)^2 \tilde{C}_{526413} = C_{35}C_{46}C_{13}C_{35}$
463512	1, 2, 3, 5, 6	$(1 + q)^2 \tilde{C}_{463512} = C_{24}C_{13}C_{46}C_{24}C_{45}$
536412	1, 3, 4, 5, 6	$(1 + q)^2 \tilde{C}_{536412} = C_{23}C_{35}C_{13}C_{46}C_{35}$
562413	1, 2, 4, 5, 6	$(1 + q)^2 \tilde{C}_{562413} = C_{35}C_{46}C_{13}C_{35}C_{23}$
563142	1, 2, 3, 4, 6	$(1 + q)^2 \tilde{C}_{563142} = C_{45}C_{24}C_{46}C_{13}C_{24}$
563412	1, 2, 3, 5, 6	$(1 + q)^2 \tilde{C}_{563412} = C_{45}C_{24}C_{13}C_{46}C_{24}C_{45}$

Conjecture 5.4 of [5] asserts that no such factorization exists when w contains the pattern 45312.

Theorem 1. *Among the 26 permutations in $\mathfrak{S}_6$ containing 45312, the element $\tilde{C}_w$ has a reversal factorization if and only if*

$$w \in \mathcal{F} := \{463152, 526413, 463512, 536412, 562413, 563142, 563412\}.$$

The seven factorizations are displayed in Table 1. In particular, Conjecture 5.4 of [5] is false.

Proof. Direct multiplication in the T -basis gives

$$(1) \quad C_{24}C_{13} = (1 + q)\tilde{C}_{431256},$$

$$(2) \quad \tilde{C}_{431256}C_{46} = \tilde{C}_{431652},$$

$$(3) \quad \tilde{C}_{431652}C_{24} = (1 + q)\tilde{C}_{463152}.$$

For the interval sequence $([2, 4], [1, 3], [4, 6], [2, 4])$, the only $I_{i,j}$ of cardinality greater than one are $I_{1,2} = I_{2,4} = \{2, 3\}$; hence $f_{\mathcal{A}}(q) = (1 + q)^2$. Equations (1)–(3) therefore give the first row of Table 1.

Two further exact identities are

$$\tilde{C}_{463152}C_{45} = \tilde{C}_{463512}, \quad C_{45}\tilde{C}_{463512} = \tilde{C}_{563412}.$$

They give the rows for 463512 and 563412. The standard anti-involution and diagram automorphism satisfy

$$\iota(\tilde{C}_w) = \tilde{C}_{w^{-1}}, \quad \omega(\tilde{C}_w) = \tilde{C}_{w_0 w w_0}, \quad w_0 = 654321,$$

where ι reverses products and $\omega(C_{ab}) = C_{7-b7-a}$. Applying these symmetries gives the remaining four rows; the three resulting orbits are

$$\{463152, 526413\}, \quad \{463512, 536412, 562413, 563142\}, \quad \{563412\}.$$

A direct inspection of the interval sequences gives $f_{\mathcal{A}}(q) = (1 + q)^2$ in every row.

It remains to exclude the other nineteen cases. Thirteen of them are

$$\mathcal{N}_1 = \{156423, 256413, 453126, 453162, 456312, 463125, 465312, \\ 516423, 546312, 564123, 564132, 564213, 564312\}.$$

Every element of $\mathcal{N}_1$ avoids 4231, contains 3412, and has $\text{gap}_{3412}(w) > 1$, where gap_{3412} is the 3412-gap of [5, Sec. 5]. The gap is 2 for the first twelve permutations and 3 for 564312. Theorem 5.2 of [5] therefore rules out reversal factorizations for all of them.

For the remaining six permutations, the following pairs satisfy $v < w$ and have the indicated Kazhdan–Lusztig polynomial:

w	v	$P_{v,w}(q)$	w	v	$P_{v,w}(q)$
356412	213456	$1 + q^2$	453612	123465	$1 + q^2$
561423	213456	$1 + q^2$	563124	123465	$1 + q^2$
564231	134526	$1 + q^2$	645312	152346	$1 + q^2$

The coefficient of q is an internal zero, so Theorem 5.3 of [5] excludes a reversal factorization in every case. The three sets above have sizes 7, 13, and 6, and exhaust the 26 permutations in $\mathfrak{S}_6$ containing 45312.

All displayed identities, pattern enumerations, gap values, and Kazhdan–Lusztig polynomials were checked with exact arithmetic in $\mathbb{Z}[q]$. The basis was generated by the standard Kazhdan–Lusztig recurrence [2, Chap. 5], and the products were evaluated from the defining Hecke relation; no numerical specialization was used. $\square$

Corollary 2. *For every $n \geq 6$, Conjecture 5.4 of [5] has a counterexample in $\mathfrak{S}_n$.*

Proof. Embed $\mathfrak{S}_6$ as the standard parabolic subgroup of $\mathfrak{S}_n$ fixing $7, \dots, n$. Compatibility of the Kazhdan–Lusztig basis with standard parabolic subgroups preserves the first factorization in Table 1, while $4631527 \cdots n$ still contains 45312. $\square$

Remark 3 (Prior work and scope). Larson [4, Ex. 6.11] constructed a small Gelfand–MacPherson resolution for $w = 463152$ with resolution data

$$I_0 = \{2, 3, 5\}, \quad I_1 = \{1, 2, 4, 5\}, \quad I_2 = \{2, 3, 5\},$$

and overlaps $J_1 = J_2 = \{2, 5\}$. Under the correspondence developed by Agrawal and Sotirov [1, Cor. 6.21 and Sec. 8], these parabolic data yield a Kazhdan–Lusztig factorization. In the present normalization, splitting the disconnected parabolics gives

$$(1 + q)^4 \tilde{C}_{463152} = C_{24} C_{56} C_{13} C_{46} C_{24} C_{56}.$$

Since C_{56} commutes with C_{13} and C_{24} , and

$$C_{56} C_{46} = C_{46} C_{56} = (1 + q) C_{46},$$

this contracts to the first identity of Table 1. Thus the geometric existence of the 463152 example is implicit in prior work. Theorem 1 records explicit

reversal-factorization certificates in the precise form of Conjecture 5.4 and gives the complete classification of its $\mathfrak{S}_6$ instances. The broader characterization problem posed in [6, Quest. 4.5] remains open.

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